

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

AUTOMOBILE ENGINEERING

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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| (a) Explain the need of crank case ventilation. | 2M |
| (b) Write down the function of piston rings. | 2M |
| (c) Define "Multi point injection" of petrol engine. | 2M |
| (d) What is the spark advance and retard mechanism? | 2M |
| (e) Draw a leaf spring. | 2M |
| (f) Write the principle of steering. | 2M |
| (g) What is meant by tandem cylinder? | 2M |
| (h) Why the shock absorbers are used in automobile? | 2M |
| (i) What is the function of a battery in any automobile? | 2M |
| (j) Which type of electrodes is used in a fuel cell? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 | Discuss about the chassis and body components in automobile. | 10M |
| OR | | |
| 3 | With the help of a neat sketch explain Splash Lubrication system? Also discuss its advantages and disadvantages. | 10M |
| 4 | Describe in detail about the multipoint fuel injection for S.I. engines. | 10M |
| OR | | |
| 5 | What is the function of a carburettor? Briefly describe the working with a neat sketch. | 10M |
| 6 | Explain the functioning of independent front wheel suspension system with torsion bar in an automobile. | 10M |
| OR | | |
| 7 | Explain the term "Backlash" in Steering Gear. Write the common procedure adopted to adjust the backlash in steering gear. | 10M |
| 8 | Explain the working principles of Hydraulic braking system with neat sketches. | 10M |
| OR | | |
| 9 | What is ABS and how does ABS helps in improving uncontrolled skidding? | 10M |
| 10 | What is a fuel cell? With a neat sketch explain its working principle. Write down its limitations. | 10M |
| OR | | |
| 11 | Discuss about the Electronic brake distribution system with suitable sketches. | 10M |
